Final Project

Perform docking from software and server to protein you interest and set of ligands from Pubchem db Then upload the complex In final docking to protein – ligand interaction profiler to have bonds profiling And finaly upload all files and screens you get to google drive link and send it on form

Form

Name: Mona Fakhri Salem

E- mail: monafakhri50@gmail.com

University: Cairo University

Faculty of Computer Science(Bioinformatics)

Affiliation

- Protein name: 6LU7

The crystal structure of COVID-19 main protease in complex with an inhibitor N3

- Protein pdb code: 6LU7

https://doi.org/10.2210/pdb6lu7/pdb

- Protein function:

The main protease of SARS-CoV-2, crucial for viral replication.

- Proteolysis: It breaks down large proteins into smaller pieces, which helps activate proteins necessary for the virus replication process.

- Controlling the virus life cycle: It plays a vital role in the formation of the virus and ensuring its continuity.

- Importance: It helps provide new therapeutic options to combat COVID-19, especially given the lack of effective treatments currently available.

- Ligands names:

Cetirizine ,N3 Inhibitor, Ebselen ,Remdesivir, DMSO (Dimethyl sulfoxide),

Nitrobenzaldehyde

-Docking tools you used:

1. **conf.txt , run.bat , vina.exe**
2. -Autodock
3. -openBable
4. Discovery Studio Visualizer

- **Databases**

1. Protein Data Bank (PDB
2. DrugBank
3. PubChem

Docking results:

Ligand	Docking results
Cetirizine	-6.0
N3 Inhibitor	-6.0
, Ebselen	-6.3
Remdesivir	-6.3
DMSO (Dimethyl sulfoxide)	-2.3
Nitrobenzaldehyde	-4.3

google drive link:

https://drive.google.com/drive/folders/1hNHiM5DUHLWB_bQN6lW_Q9bxmc023MZp?usp=drive_link

LINK2

https://drive.google.com/drive/folders/1225_JltrzBe65I99sLX6p0RbxZ_DOxLk?usp=drive_link